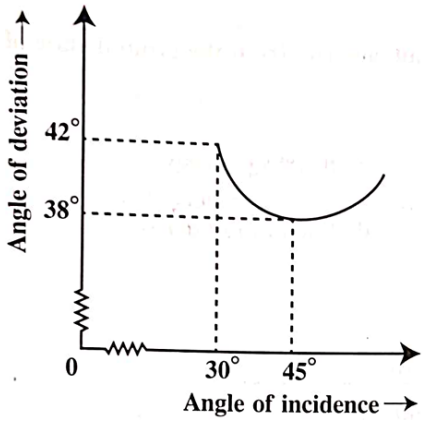
GOVERNMENT OF KARNATAKA
KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD
II PUC EXAM – 2, MAY 2026
SCHEME OF EVALUATION

SUBJECT: 33 - PHYSICS **MAX. MARKS: 70**

Q. No.	DESCRIPTION	MARKS
	PART – A	
	I. Pick the correct option among the four given options for ALL of the following questions: **15 × 1 = 15**	
1	**Which of the following is not a basic property of electric charge?** a) Charge is quantised b) Charge is conserved **c) Charge is vector** d) Charge is additive **ANS: c) Charge is vector**	**1**
2	**The values of electric field (E) and electric potential (V) at any point on the equatorial plane of an electric dipole are such that** a) E = 0 and V = 0 **b) E ≠ 0 and V = 0** c) E = 0 and V ≠ 0 d) E ≠ 0 and V ≠ 0 **ANS: b) E ≠ 0 and V = 0**	**1**
3	**A Wheatstone bridge is said to be balanced only if** **a) The current through the galvanometer is zero.** b) All the four resistances in the bridge must be equal. c) The current through the galvanometer is non-zero. d) The potential difference across the galvanometer is non-zero. **ANS: a) The current through the galvanometer is zero.**	**1**
4	**If the number of turns in a moving coil galvanometer is doubled, then** a) both current sensitivity and voltage sensitivity doubles b) both current sensitivity and voltage sensitivity remain same **c) current sensitivity doubles and voltage sensitivity remains same** d) current sensitivity remains same and voltage sensitivity doubles **ANS: c) current sensitivity doubles and voltage sensitivity remains same**	**1**
5	**Consider the following statements:** **Statement I : The net magnetic flux through any closed surface is zero.** **Statement II : The number of magnetic field lines leaving a closed surface is balanced by the number of magnetic field lines entering the surface.** **Among the above two statements,** a) Both the statements are false **b) Both the statements are true** c) Statement I is true and Statement II is false d) Statement I is false and Statement II is true **ANS: b) Both the statements are true**	**1**

6	**Lenz's law is the consequence of law of conservation of _____** **a) Energy** b) Charge c) Mass d) Momentum **ANS: a) Energy**	**1**
7	**When an alternating voltage is applied to a resistor, which of the following statements is wrong?** a) The average current over a complete cycle is zero b) The average voltage over a complete cycle is zero c) The phase difference between voltage and current is zero **d) The average power dissipated by the resistor over a complete cycle is zero.** **ANS: d) The average power dissipated by the resistor over a complete cycle is zero.**	**1**
8	**Which of the following electromagnetic waves are suitable for RADAR systems used in aircraft navigation?** **a) Microwaves** b) X-rays c) Ultraviolet rays d) Gamma rays **ANS: a) Microwaves**	**1**
9	**The graph plotted between angle of deviation versus angle of incidence for a triangular prism is as shown below. What is the angle of minimum deviation?** a) 30° **b) 38°** c) 42° d) 45° **ANS: b) 38°**  	**1**
10	**When compared to the eyepiece, the objective lens of refracting telescope must have** **a) Larger focal length and larger aperture** b) Larger focal length and smaller aperture c) Smaller focal length and larger aperture d) Smaller focal length and smaller aperture **ANS: a) Larger focal length and larger aperture**	**1**
11	**Two light sources are said to be coherent, when both the sources emit light of** a) Same speed b) Same intensity c) Same amplitude **d) Same wavelength and constant phase difference.** **ANS: d) Same wavelength and constant phase difference.**	**1**
12	**An electron (e^-), a proton (p), a neutron (n) and an alpha (α) particle are moving with same speed. Which of these particles has the shortest de Broglie wavelength?** **(Given: Masses are $m_e < m_p < m_n < m_\alpha$)** a) Electron b) Proton c) Neutron **d) Alpha particle** **ANS: d) Alpha particle**	**1**

13	**The minimum energy required to free an electron from the ground state of hydrogen atom is called_____** a) Work function b) Binding energy **c) Ionization energy** d) Excitation energy **ANS: c) Ionization energy**	**1**
14	**Match the following:** **Column – I** | **Column – II** **(i) Isotopes** | **(p) $_1^3$H and $_2^4$He** **(ii) Isobars** | **(q) $_1^1$H and $_1^2$H** **(iii) Isotones** | **(r) $_1^3$H and $_2^3$He** a) (i)→ (r), (ii) → (p), (iii) → (q) **b) (i)→ (q), (ii)→(r), (iii) → (p)** c) (i)→(p), (ii)→ (q), (iii) → (r) d) (i) → (p), (ii)→(r), (iii) → (q) **ANS: b) (i)→ (q), (ii)→(r), (iii) → (p)**	**1**
15	**Which of the following elements has to be doped with silicon to obtain n-type semiconductor?** a) Aluminium b) Boron **c) Phosphorus** d) Indium **ANS: c) Phosphorus**	**1**

II. Fill in the blanks by choosing appropriate answer from the given options for all the following questions: **5 × 1 = 5**

[Diffraction, Magnetisation, Polarisation, AC generator, Photon, Transformer]

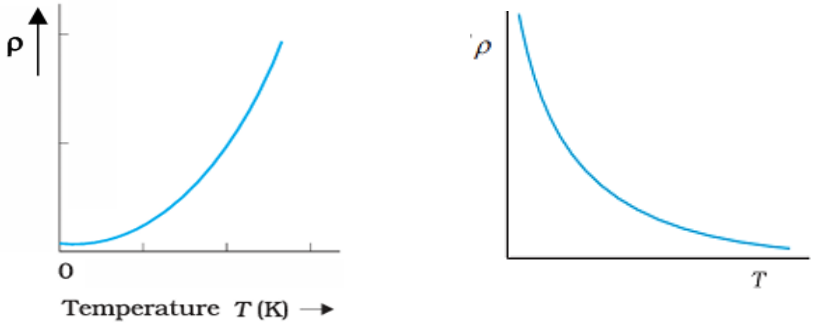
16	The net magnetic moment per unit volume of a sample is called **Magnetisation.**	**1**
17	The device which converts mechanical energy into electrical energy is **AC generator.**	**1**
18	The device which works on the principle of mutual induction is **Transformer.**	**1**
19	The bending of light around the sharp edges and its entry into expected geometrical shadow is called **Diffraction.**	**1**
20	The tiny packet of energy of light is called **Photon.**	**1**

PART – B

III. Answer any FIVE of the following questions: **5 × 2 = 10**

21	**Two identical point charges are separated by a distance 0.2 m in air repel each other with a force 9×10^9 N. Find the magnitude of each charge. (Given $\frac{1}{4\pi\varepsilon_0} = 9\times10^9$ Nm²/C²)**	
ANS	Electrostatic force: $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$ Both charges are identical, let the charges be $q_1 = q_2 = q$ and $r = 0.2$ m $9\times10^9 = 9\times10^9 \times \frac{q^2}{0.2^2} \Rightarrow q = 0.2$ C	**1** **1**
22	**Name any two factors on which the capacitance of a parallel plate capacitor depends.**	
ANS	(i) Area of the plates (ii) Distance between the plates (iii) Dielectric constant of the medium between the plates. **(ANY TWO)**	**1** **1**

23 **Graphically represent the variation of resistivity with absolute temperature for the following materials: i) Copper ii) Silicon** — **1+1**



24 **Obtain an expression for the radius of circular path described by a charged particle moving perpendicular to the uniform magnetic field.**

ANS

The magnetic force acts as the centripetal force. $F_B = F_C \Rightarrow q\,v\,B = \frac{m\,v^2}{r}$ — **1**

$\Rightarrow$ Radius of the circular path: $r = \frac{m\,v}{q\,B}$ — **1**

25 **How does the self inductance of a solenoid change if**
i) an iron rod is inserted into it? ii) current through it is increased?

ANS

i) Self-inductance **increases.** — **1**

ii) Self-inductance **remains same (no change).** — **1**

26 **What is displacement current? Write the mathematical form of Ampere-Maxwell law.**

ANS

The current due to changing electric flux (or electric field) with time is called displacement current.

OR The current due to time varying electric flux (or electric field) is called displacement current. — **1**

Ampere-Maxwell law: $\oint \vec{B} \cdot \vec{dl} = \mu_0\, i_c + \mu_0 \varepsilon_o \left(\frac{d\phi_E}{dt} \right)$ — **1**

27 **Give two limitations of Bohr's model of hydrogen atom.**

ANS

(i) Bohr's model is applicable only to hydrogenic (single electron) atoms.

OR Bohr's model is not applicable to atoms with two or more electrons. — **1**

(ii) This model is unable to explain the relative intensities of the frequencies emitted by hydrogenic (hydrogen like) atoms. **(ANY OTHER CORRECT ANSWER)** — **1**

28 **Distinguish between intrinsic and extrinsic semiconductors.**

ANS

Intrinsic semiconductors	**Extrinsic semiconductors**
Semiconductor without doping (pure form)	Semiconductor doped with impurity atoms (impure form)
Number of free electrons (n_e) is equal to the number of holes (n_h), or $n_e = n_h$	Number of free electrons (n_e) and number of holes (n_h) are unequal or $n_e \neq n_h$
Conductivity is low.	Conductivity is high.
Conductivity depends only on temperature.	Conductivity depends both on temperature and impurity added.
Conductivity is equally due to electrons and holes.	Conductivity is mainly due to majority charge carriers.

(ANY TWO – Any other correct difference) — **1 + 1**

PART – C

IV. Answer ANY FIVE of the following questions: **5 × 3 = 15**

29 **Derive an expression for the torque experienced by an electric dipole placed in a uniform electric field.**

ANS

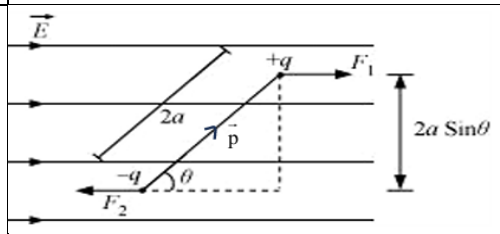


Diagram **1**

Consider an electric dipole of dipole moment $\vec{p}$ is placed in a uniform external electric field $\vec{E}$.

Magnitude of torque = force × perpendicular distance (Force on q is , F = qE) **1**

$\tau = q\,E \times (2a \sin\theta) = (2aq)\,E \sin\theta$; ($\because$ 2aq = p, dipole moment)

$\tau = p\,E \sin\theta$ **1**

30 **List out the three important results regarding electrostatics of conductors.**

ANS

a) The interior of a conductor can have no excess charge in the static situation. **1**

b) Inside a conductor, electrostatic field is zero. **1**

c) On the surface of a charged conductor, electrostatic field must be normal to the surface at every point. **1**

d) Electrostatic potential is constant throughout the volume of the conductor and has the same value (as inside) on its surface.

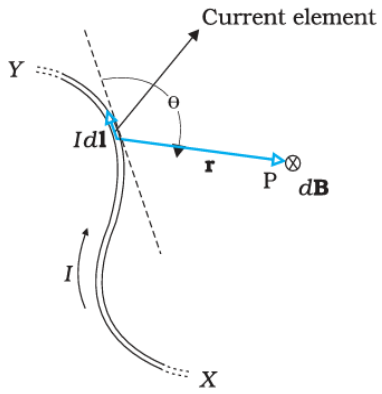
e) Electric field at the surface of a charged conductor, $\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$; where σ is the surface charge density and $\hat{n}$ is a unit vector normal to the surface **(ANY THREE)**

31 **State and explain Biot-Savart law with a relevant diagram.**

ANS

Statement: The magnitude of the magnetic field dB due to a current element is proportional to the current I, element length d*l*, and inversely proportional to the square of the distance r. **1**

Diagram **1**



Magnetic field: $dB = \frac{\mu_0}{4\pi} \times \frac{I\,dl\,\sin\theta}{r^2}$ **1**

OR $dB = \frac{\mu_0}{4\pi}\left(\frac{I\left|d\vec{l} \times \vec{r}\right|}{r^3}\right)$

OR $d\vec{B} = \frac{\mu_0}{4\pi}\left(\frac{I\,d\vec{l} \times \vec{r}}{r^3}\right)$

OR $d\vec{B} = \frac{\mu_0}{4\pi}\left(\frac{I\,d\vec{l} \times \hat{r}}{r^2}\right)$ **OR** $dB = \frac{\mu_0}{4\pi}\frac{I\left|d\vec{l} \times \hat{r}\right|}{r^2}$

32 **Write the three differences between diamagnetic and ferromagnetic materials**

ANS

Diamagnetic Materials	Ferromagnetic Materials	
(i) These are the materials that are weakly repelled by a magnetic field.	(i) These are the materials that are strongly attracted by a magnetic field.	**1**
(ii) Magnetic susceptibility value for them is negative. (**OR** $\chi < 0$)	(ii) Magnetic susceptibility value for them is positive. (**OR** $\chi \gg 0$)	**1**
(iii) Relative permeability values for them have ranges $0 \le \mu_r < 1$. **OR** Relative permeability is less than 1.	(iii) Relative permeability values for them have ranges $\mu_r \gg 1$. **OR** Relative permeability is more than 1.	**1**

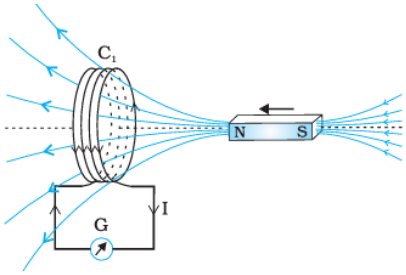
(iv) There is no formation of domains.	(iv) There is formation of domains.	
(v) Atoms of these materials do not have net magnetic moment.	(v) Atoms of these materials have net magnetic moment.	
(vi) When placed in an external magnetic field, the magnetic field inside these materials is much less than the value of the field outside.	(vi) When placed in an external magnetic field, the magnetic field inside these materials is very much greater than the value of the field outside.	
(vii) Magnetic field lines are expelled out, when the diamagnetic material is placed in an external magnetic field.	(vii) Magnetic field lines enter inside when the ferromagnetic material is placed in an external magnetic field.	

(ANY THREE; Any other correct difference has to be considered)

33 **Describe the coil and bar magnet experiment to demonstrate the phenomenon of electromagnetic induction**

ANS

Diagram



1

- When the North-pole of a bar magnet is pushed towards the coil, the pointer in the galvanometer deflects. **1**
- The galvanometer does not show any deflection when the magnet is held stationary. **1**
- When the magnet is pulled away from the coil, the galvanometer shows deflection in the opposite direction, which indicates reversal of the current's direction.
- When the South-pole of the bar magnet is moved towards or away from the coil, the deflections in the galvanometer are opposite to that observed with the North-pole for similar movements.
- The deflection is found to be larger when the magnet is pushed towards or pulled away from the coil faster.
- When bar magnet is held fixed and the coil C_1 is moved towards/away from the magnet, same effects are observed
- The relative motion between the magnet and the coil is responsible for generation (induction) of electric current in the coil. **(ANY TWO OBSERVATIONS)**

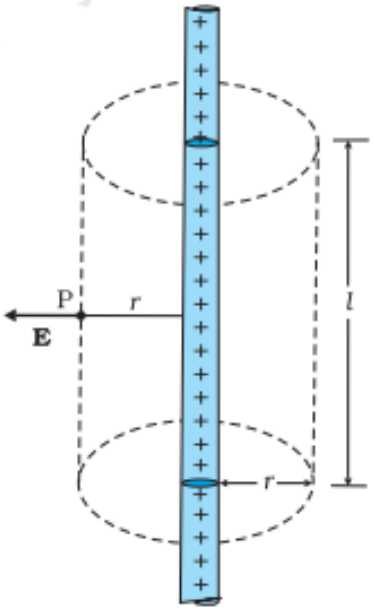
34 **Give the three Cartesian sign conventions used in spherical mirrors.**

- All the distances are measured from the pole of the mirror. **1**
- The distances measured in the same direction as the incident light are taken as positive and the distances measured in the direction opposite to the direction of incident light are taken as negative. **1**
- The heights measured upwards (normal to the principal axis of the mirror) are taken as positive. The heights measured downwards are taken as negative. **1**

35	**The work function of Caesium is 2.14 eV. When light of frequency 6×10^{14} Hz is incident on Caesium surface, find the maximum kinetic energy of photoelectrons emitted. (Given : Planck's constant, $h = 6.63 \times 10^{-34}$ J s and $e = 1.6 \times 10^{-19}$ C)**	
ANS	Maximum kinetic energy of photoelectrons : $K_{max} = h\upsilon - \phi_0$ $K_{max} = 6.63 \times 10^{-34} \times 6 \times 10^{14} - 2.14 \times 1.6 \times 10^{-19}$ $K_{max} = 3.98 \times 10^{-19} - 3.42 \times 10^{-19}$ $K_{max} = 0.56 \times 10^{-19}$ J	**1** **1** **1**
OR	Maximum kinetic energy of photoelectrons : $K_{max} = h\upsilon - \phi_0$ $K_{max} = \frac{6.63 \times 10^{-34} \times 6 \times 10^{14}}{1.6 \times 10^{-19}} - 2.14$ $K_{max} = 2.49 - 2.14 = 0.35$ eV **(ANY OTHER CORRECT METHOD)**	**1** **1** **1**
36	**Mention the three features of nuclear force.**	
ANS	(i) Nuclear force is strongest force. (ii) Nuclear force is short ranged. (iii) Nuclear force shows the property of saturation. (iv) Nuclear force does not depend on the electric charge (charge independent). **(ANY THREE - Any other correct feature to be considered)**	**1** **1** **1**

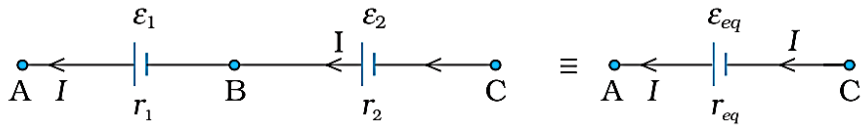
PART – D

V. Answer any THREE of the following questions: 3 × 5 = 15

37.	**State Gauss's law in electrostatics. Using it, arrive at the expression for the electric field due to an infinitely long straight uniformly charged conducting wire.**	
ANS	 **Statement:** The electric flux through any closed surface is equal to $\frac{1}{\varepsilon_0}$ times the total (net) charge enclosed by the surface S. **Diagram** Consider an infinitely long uniformly charged straight wire with uniform linear charge density λ. Let us calculate the electric field at point P at a distance r from this wire. We will consider a cylindrical Gaussian surface of radius r and length l as shown in the figure. By definition, electric flux through the Gaussian cylinder is $\phi_E = \int \vec{E} \cdot d\vec{A} = E \times 2\pi rl$ .....**(1)** By Gauss's law in electrostatics, flux though the closed surface is $\phi_E = \frac{q}{\varepsilon_o}$ and charge enclosed: $q = \lambda l$ $\Rightarrow \phi_E = \frac{\lambda\, l}{\varepsilon_o}$ .....**(2)** Comparing the two equations (1) and (2), $E \times 2\pi rl = \frac{\lambda\, l}{\varepsilon_o}$ $\Rightarrow E = \frac{\lambda}{2\pi\varepsilon_o r}$	**1** **1** **1** **1** **1**

38 **Obtain the expressions for equivalent emf and equivalent internal resistance, when two different cells are connected in series.**

ANS Let $\varepsilon_1, \varepsilon_2$ be the emfs of two cells and r_1, r_2 be their internal resistances respectively and they are joined in series as shown in the figure. Let the equivalent emf be ε_{eq} and equivalent internal resistance be r_{eq}. Let the terminal potential differences for the first cell be V_1, for the cell be V_2 and for combination be V.



Both figures with arrows for current — 1

For the first cell, $V_1 = \varepsilon_1 - I\,r_1$ and for the second cell, $V_2 = \varepsilon_2 - I\,r_2$. — 1

As the cells are in series, $V = V_1 + V_2$.

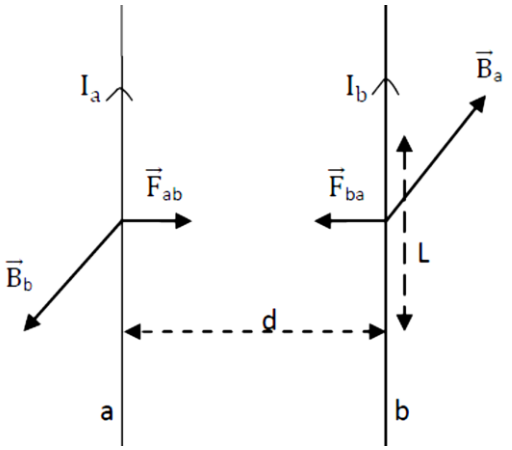
$$\Rightarrow V = (\varepsilon_1 - I\,r_1) + (\varepsilon_2 - I\,r_2) = (\varepsilon_1 + \varepsilon_2) - I(r_1 + r_2)$$ — 1

For the equivalent cell, $V = \varepsilon_{eq} - I\,r_{eq}$ — 1

By comparing the last two equations, we get: $\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$ and $r_{eq} = r_1 + r_2$ — 1

39 **Derive an expression for the force per unit length between two infinitely long straight parallel current carrying conductors placed in vacuum. Hence define the unit "ampere".**

ANS



Labelled diagram with currents, field and force. — 1

($\vec{F}_{ab}$ and $\vec{B}_b$ are not necessary in the diagram)

The magnetic field at the location of 'b' produced by the conductor 'a' is, $B_a = \frac{\mu_0 I_a}{2\pi d}$ .....(1) — 1

The magnetic force on a segment L of the conductor 'b' due to 'a' is $F_{ba} = B_a I_b L \sin 90^o$

Using equn(1), $F_{ba} = \frac{\mu_0 I_a I_b L}{2\pi d}$ — 1

This force F_{ba} is towards conductor 'a'.

Force per unit length on the conductor 'b' is $\frac{F_{ba}}{L} = \frac{\mu_0}{4\pi}\left(\frac{2I_aI_b}{d}\right) = \frac{\mu_0 I_a I_b}{2\pi d}$

Similarly, force per unit length on the conductor 'a' is $\frac{F_{ab}}{L} = \frac{\mu_0 I_a I_b}{2\pi r}$ and $\vec{F}_{ab} = -\vec{F}_{ba}$ — 1

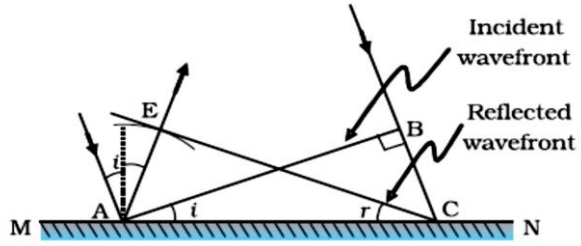
Definition of ampere: **One ampere** is that steady **current** which, when maintained in each of **two infinitely long, straight, parallel conductors** of negligible cross-section and placed **one metre apart** in **vacuum (or free space or air)**, would produce a **force of 2 × 10^{-7} newton per metre** of length on each other. — 1

(Note: Any other equivalent correct definition with required key terms should be considered)

40 **i) Using Huygen's Principle, show that the angle of incidence is equal to the angle of reflection when a plane wavefront is reflected by a plane surface. (3)**

ii) Mention two uses of polaroids. (2)

ANS i)



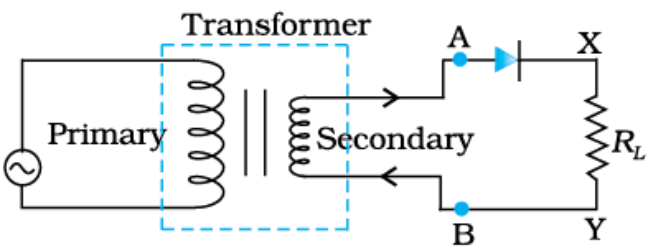
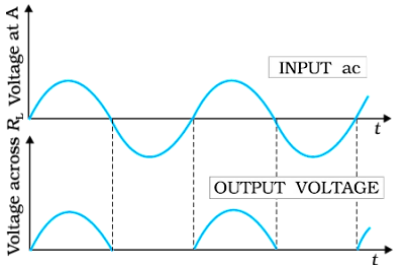
Labelled diagram — 1

MN → Reflecting plane surface (mirror)

i → angle of incidence

r → angle of reflection

v → speed of light wave

	Let τ be the time taken by the wave front to travel from the point B to C, then distance BC= vτ. In order the construct the reflected wave front, a sphere of radius = vτ, is drawn from the point A as shown in the above figure. Let CE represent the tangent plane drawn from the point C to this sphere. Then **AE = BC = vτ , ∠ABC= ∠CEA = 90° and AC is common.** **1** **The triangles ABC and CEA are congruent. Thus, i = r** **1**	
ii)	ii) Uses of polaroids: sunglasses, windowpanes, photographic cameras, 3D movie cameras **(ANY TWO – Any other correct use)**	**1+1**
41	**i) What is a rectifier? (1)** **ii) Draw the circuit diagram and input-output waveforms for half-wave rectifier. (2)** **iii) Explain the working of half-wave rectifier. (2)**	
ANS	**i)** A rectifier is a circuit (device) which converts AC into DC. **1** **ii)** **Circuit diagram** **Waveform** **1+1**   **iii) Working:** During positive half cycle of AC, A is positive and B is negative. The diode becomes forward biased and it conducts. The output appears across the resistance R_L. **1** During negative half cycle of AC, A becomes negative and B becomes positive. The diode is reverse biased, and it does not conduct. No output appears in this case. **1**	
VI. Answer any TWO of the following questions:		**2 × 5 = 10**
42	**Three point charges +4μC, –6μC and +8μC are placed at the corners A, B and C respectively of a square ABCD of side 10 cm. Calculate the work done to transfer a point charge +3nC from the corner D to the centre O of the square. (Given $\frac{1}{4\pi\varepsilon_0} = 9\times10^9$)**	
ANS	Given: $q_1 = +4\mu C = 4\times10^{-6}$ C , $q_2 = -6\ \mu C = -6\times10^{-6}$ C and $q_3 = +8\mu C = 8\times10^{-6}$ C AD = CD = 10 cm = 0.1 m, diagonal BD = $0.1\sqrt{2}$ m, AO = BO = CO = $\frac{0.1\sqrt{2}}{2} = 0.05\sqrt{2}$ m Electric potential at a distance r from a point charge q is $V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$ **1** Electric potential at O is, $V_O = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{AO}+\frac{q_2}{BO}+\frac{q_3}{CO}\right] = 9\times10^9\left[\frac{4-6+8}{0.05\sqrt{2}}\right]\times10^{-6} = 764\times10^3$ V **1** Electric potential at D is , $V_D = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{AD}+\frac{q_2}{BD}+\frac{q_3}{CD}\right] = 9\times10^9\left[\frac{4}{0.1}+\frac{-6}{0.1\sqrt{2}}+\frac{8}{0.1}\right]\times10^{-6} = 698\times10^3$ V **1** Work done to transfer a charge of 3×10^{-9} C from D to O is $W = q(V_O - V_D)$ **1** $W = 3\times10^{-9}\times(764\times10^3 - 698\times10^3) = 198\times10^{-6}$ J **OR** 1.98×10^{-4} J **1** **(Any other correct alternative method should be considered)**	
43	**A copper wire of length 50 cm and cross-sectional area 0.3 mm^2 carries a current of 1.5A. The number density of free electrons in copper is 8.5 × 10^{28} m^{-3}. Determine the drift velocity of free electrons and the current density in the wire. (Given: e=1.6×10^{-19}C)**	

ANS	Current: $I = n\,e\,A\,v_d$ OR Drift velocity : $v_d = \frac{I}{neA}$	**1**
	$v_d = \frac{1.5}{8.5\times10^{28}\times1.6\times10^{-19}\times0.3\times10^{-6}}$	**1**
	$v_d = 0.368\times10^{-3}\ \text{m s}^{-1}$	**1**
	Current density: $j = \frac{I}{A}$	**1**
	$j = \frac{1.5}{0.3\times10^{-6}} = 5\times10^{6}\ \text{Am}^{-2}$	**1**
	(Any other correct alternative method should be considered)	
42	**A series LCR circuit with L = 0.5 H, C = 8 μF, R = 20 Ω is connected to a variable frequency 220 V AC source.** **i) Determine the source frequency that drives the circuit into resonance.** **ii) Obtain the impedance and amplitude of current at resonance.**	
ANS	**i)** Resonant Frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$ OR f_o or $\upsilon_0 = \frac{1}{2\pi\sqrt{LC}}$	**1**
	$\omega_0 = \frac{1}{\sqrt{0.5\times8\times10^{-6}}} = 500\ \text{rad s}^{-1}$ | $\upsilon_0 = \frac{1}{2(3.14)\sqrt{0.5\times8\times10^{-6}}} = 80\ \text{Hz}$	**1**
	ii) Impedance at resonance: $Z = R = 20\ \Omega$	**1**
	RMS value of current at resonance: $I = \frac{V}{R} = \frac{220}{20} = 11\ \text{A}$	**1**
	Amplitude of current at resonance: $i_m = I\sqrt{2} = 11\times1.414 = 15.6\ \text{A}$	**1**
	(Any other correct alternative method should be considered)	
45.	**The radii of curvature of two surfaces of a convex lens are 20 cm and 22 cm.** **Find the focal length and power of the lens.** **(Given: Refractive index of material of lens is 1.5).**	
ANS	$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ **OR** $\frac{1}{f} = (n_{21}-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$; where $n_{21} = \frac{n_2}{n_1} = \frac{1.5}{1} = 1.5$	**1**
	Substitution of the given values, $\frac{1}{f} = (1.5-1)\left(\frac{1}{20} - \frac{1}{-22}\right)$	**1**
	$\Rightarrow \frac{1}{f} = (0.5)\left(\frac{1}{20} + \frac{1}{22}\right)$	
	Solving, focal length: f = 20.95 cm ≈ 21 cm **OR** 0.21 m	**1**
	Power of the lens : $P = \frac{1}{f}$	**1**
	$P = \frac{1}{0.21} = 4.76 \approx 4.8\ \text{D}$	**1**
	(Any other correct alternative method should be considered)	
VII.	**PART – E** **(For Visually-challenged Students only)**	
9)	**When the angle of incidence of a ray of light in a triangular prism is increased, how does the angle of deviation change?** a) Increases b) Decreases c) Remains same **d) First decreases then increases.** **ANS: d) First decreases then increases.**	**1**

Note: Any other alternate correct method/answer should be considered.